Nuclear-spin memory across turnovers of a cryptochrome-type radical pair and its relaxation limit at the geomagnetic field

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The radical-pair mechanism turns the spin state of a short-lived pair into product yields and nuclear polarisation of the products. Both are understood within one reaction cycle; what a pair regenerated on the same molecule carries from one cycle to the next has not been quantified. Here we show that the nuclear spins, which outlive the pair in its diamagnetic product, hold a memory of earlier cycles, which we measure with the capacity measures of reservoir computing. Solving the spin dynamics of a flavin–tryptophan pair with cryptochrome's couplings at 50 μT, we find that an end-point yield carries no memory, time-resolved yields carry some, and the product's nuclear polarisation carries the most. The memory resides in populations of the product's protons, so its lifetime is their T_1, predicted at the geomagnetic field to be milliseconds, not the seconds measured at high field. This gives a criterion with three measurable constants: the turnover interval, the low-field proton T_1, and the number of molecules read as one pool. It bounds the conjecture that radical-pair chemistry computes in neurons: the memory lasts one to three milliseconds unless the register's local motion is several times faster than protein tumbling.

I. INTRODUCTION

A magnetic field as weak as the Earth's, about 50 μT, can change the outcome of a chemical reaction through the radical-pair mechanism[1,2,16]: two radicals born with correlated electron spins interconvert coherently between singlet and triplet under the hyperfine fields of nearby nuclei, and spin-selective recombination turns that interconversion into product yields. The same spin sorting leaves the nuclei of the products polarised (chemically induced dynamic nuclear polarisation, CIDNP[18]), as observed in flavoproteins[40,53]. The mechanism underlies the leading model of avian magnetoreception, in which cryptochrome hosts a flavin–tryptophan radical pair[2–5,12]; only a spatially separated pair, with a microsecond lifetime and weak electron–electron coupling, keeps the coherence that weak-field sensitivity requires[23].

Both effects are understood as the physics of one reaction cycle: a pair is born, evolves and recombines. Many radical-pair reactions, however, are cyclic. The same flavin is reduced and reoxidised turnover after turnover, and the nuclei are not consumed by recombination: they leave in the diamagnetic product, polarised, and relax far more slowly than the pair lives. The spin state in which one cycle begins therefore depends on the cycles before it. Wong *et al.*[35] asked whether nuclear polarisation accumulates over cryptochrome photocycles enough to help the compass. What has not been quantified is the general property behind that question: how much information about earlier cycles the products of the current cycle carry, in which observable, and for how long.

The question has a second motivation. The success of the mechanism has repeatedly prompted a bolder conjecture, that spin coherence in biomolecules might support not merely sensing but information processing in a nervous system[6,7], contested on decoherence grounds[9] and part of the broader programme of quantum biology[10,11]. Any such proposal needs exactly this quantity, the information one chemical cycle hands to the next, and none has named the observable a downstream reaction would read, the clock on which it is read, or the relaxation constant that bounds it.

This paper computes it. We ask one question: *how much does a cryptochrome-type radical pair, regenerated repeatedly on the same molecule, remember of its earlier cycles, and can ordinary chemistry read that memory back?* We simulate the pair's spin dynamics, with recombination and decoherence, through many successive cycles driven by different inputs, and ask how much of the past input sequence the chemical products of the current cycle still encode. The measure is borrowed from reservoir computing[13,24,25], in which a fixed dynamical system is driven by a time series and a linear readout of its state is trained to reconstruct past inputs; it serves here as an instrument for measuring memory, not as a claim that the molecule computes. Its requirement that old inputs be eventually forgotten (fading memory) is supplied by the openness of the chemical cycle, which discards the electron pair every turnover and keeps only the nuclei. What is tested is retention and numerical recovery of past inputs from the products; a reaction that would use the recovered quantity in a living cell is not modelled. Table I translates each term into the chemistry it stands for, and Fig. 1 shows the cycle.

Three results follow. First, the memory exists and its size depends on what is read: an end-point yield carries none, the time course of a product carries some, and the nuclear polarisation the product carries away carries the most, well above its memoryless floor. Second, the memory is held in nuclear populations and survives recom-

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bination, so its clock is the interval between turnovers, not the pair lifetime; what limits it is the *register-reuse criterion*: the same nuclear register must be reused from one turnover to the next and must survive the interval. Third, that register can only be protons, and their relaxation at the geomagnetic field, predicted from measured constants, takes milliseconds, which caps the memory horizon at one to three milliseconds. A classical network matched to the same readout performs as well, and at ambient light a photochemical cycle is far too slow to meet the criterion.

That a nuclear memory is bounded by nuclear relaxation is not surprising; what the calculation supplies is what an estimate does not. Whether any memory is observable depends on the readout: the end-point yield, the observable usually considered, has none. The memory is in populations, not coherences, and in protons only, the flavin ^{14}N being wiped by quadrupolar relaxation within a cycle. And the relevant T_1 is not the second-long value familiar from high-field NMR: at 50 μT it is more than three orders of magnitude shorter, which moves the horizon from seconds to milliseconds.

For the neural conjecture the result is a bound, drawn in Sec. IV: neural timescales are reached only if the proton register reorients four to fifteen times faster than the protein tumbles and the pair is regenerated catalytically on the same molecule. Whether a neuronal cryptochrome forms a separated radical pair at all is unestablished (mammalian CRY1/CRY2 are primarily circadian transcriptional regulators[15]), so that reading is conditional; the spin-dynamics results do not depend on it. The question arose from a layered proposal[19,20] of a nuclear-spin memory read through a radical-pair interface, of which this paper is the quantitative test. Section II describes the model and the measures, Sec. III presents the results in the order above, Sec. IV states what would test the criterion and what it implies for neurons, and the appendix gives the equations.

TABLE I. The correspondence used throughout: each term of reservoir computing, what it is in the radical pair, and how it is measured here.

term	in the radical pair	measured as
input	the singlet character of the pair at birth (alternatives: field magnitude, recombination rate)	one value per chemical cycle
reservoir state	the nuclear spins (the *register*), which outlive the electron pair	nuclear density matrix handed from one cycle to the next
readout	quantities a downstream reaction could use: product yields over time; the nuclear polarisation the product carries away	linear fit from these to past inputs
memory capacity	how many past inputs, and functions of them, the readout reconstructs	sum of squared out-of-sample correlations
memoryless floor	what the same readout scores with the register wiped every cycle	same estimator, register randomised
excess	capacity above the floor: the memory itself	difference of the two
clock	the interval between successive chemical cycles	T_{turnover}, not the pair lifetime τ
register lifetime	the time over which a nucleus loses its polarisation (T_{1n}); its coherence lifetime is T_{2n}	predicted from measured couplings at 50 μT
memory horizon	capacity above the zero-delay term, times the clock	$(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$

II. THE MODEL IN BRIEF

The spin system. Five spin-1/2 particles: the two electrons and three nuclei (two ^{14}N on the flavin, one proton on the tryptophan) with cryptochrome's couplings (supplementary material, Sec. S13), under the Haberkorn master equation[17,30] with spin-selective recombination and electronic dephasing (Appendix A). The nitrogens are propagated as spin-1/2 *within* a cycle, a computational approximation the supplementary material shows to be conservative (Sec. S11); their fast quadrupolar relaxation *between* cycles is imposed separately, by wiping them, where the criterion requires it. The *cryptochrome point* ($B = 50$ μT, lifetime $\tau = 1$ μs, no electron–electron coupling as the reference premise) is where every chemical readout, the clock and the criterion are evaluated; an *engineered point* ($B = 1$ mT, $\tau = 50$ ns, stronger couplings) is a molecular-spin reference used only for the full-readout capacity and the classical comparison.

The cycle. Each cycle the nuclei are retained and a *new* spin-correlated electron pair is born whose singlet character $s \in [0,1]$ encodes the input, $\rho_e(s) = s\,|S\rangle\langle S| + (1-s)P_T/3$; the pair evolves for its lifetime, recombines spin-selectively, and hands its nuclear register to the next cycle. The single-electron reset of earlier spin-reservoir treatments[26], which overwrites one electron with the input state, is not a chemical process and leaves a separated pair with no capacity (supplementary material, Sec. S3). The state is an ensemble density matrix, and at τ the ensemble has split into molecules that have recombined and molecules that have not; which nuclear state is handed on is a modelling choice: that of the

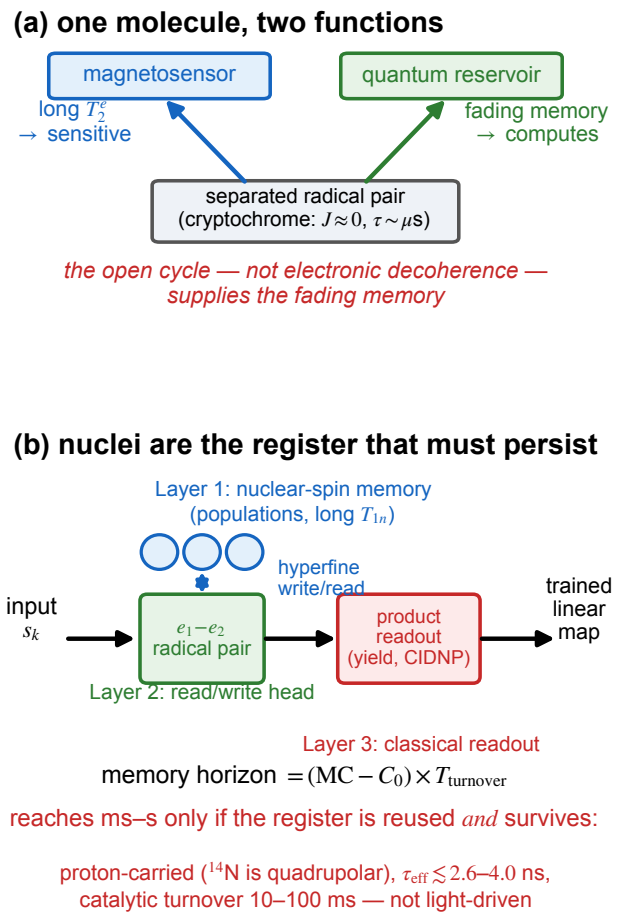


FIG. 1. **The chemical cycle as a reservoir.** (a) A cryptochrome-type separated radical pair is one molecule in two roles: its long electronic coherence makes it a weak-field magnetosensor, and the dissipation of an open cycle supplies the fading memory a reservoir needs. (b) The nuclei are the register: they survive recombination and persist from one turnover to the next, whereas the electron pair is created afresh each cycle, so the memory horizon $(\mathrm{MC}-C_0)\times T_{\mathrm{turnover}}$ is set by the turnover interval, not by the pair lifetime τ. It reaches the neural band only if the register is reused *and* survives the pause, which requires protons that reorient with $\tau_c \lesssim 2.6$–4.0 ns (supplementary material, Secs. S9 and S12).

unreacted molecules (the *survivor* register) or that carried away by the recombined ones (the *product* register, Eq. (A4)). The product register is the physically faithful one and gives larger capacities, so the survivor numbers reported are conservative. Tables II, III and IV use the survivor register; the criterion's own numbers (Table V) use the product register; numbers are tagged wherever the two differ.

The measures, concretely. A run is 700 cycles with inputs $s_1, \ldots, s_{700}$ drawn at random, after each of which the readout values (yields at five times within the cycle, and the polarisation of each product nucleus) are recorded. For each delay d we find, on the first half of the run, the weighted sum of readout values that best predicts s_{k-d}, and score it on the second half by the fraction of variance it explains, a number between 0 and 1. The linear memory capacity MC[31] is the sum of these scores over d, so MC = 2.9 means the readout carries nearly three delays' worth of input; the information processing capacity IPC[32] adds the same scores for s_{k-d}^2 and for products of two past inputs. A readout with no memory still predicts the present input ($d = 0$) and its square, so every route has a memoryless floor, measured by wiping the register every cycle; only the excess above it is memory. A concrete case fixes the scale. For the polarisation readout at the cryptochrome point the scores by delay are 1.00, 0.85, 0.47, 0.18, 0.04 for $d = 0, \ldots, 4$: the present input is recovered exactly, the previous one to 85%, the one before to 47%, and little beyond. Their sum over $d \geq 1$, 1.82, is the memory; with a 10 ms turnover the horizon, 1.82×10 ms $= 18$ ms, is this sum converted to time, a weighted total of recovery and not the age of the oldest input recovered, which here is two to three turnovers. The same readout's IPC adds the scores for s_{k-d}^2 and for products such as $s_k s_{k-1}$, and its floor of 2.00 is the present input's s and s^2 alone. The weights are fitted by us, not by the molecule. Three statements are therefore kept apart throughout: that the product *carries* information about past inputs (computed here); that a linear fit on the product's observables *recovers* it (computed here, by us); and that a downstream reaction could *implement* such a fit (not modelled; the counting-noise and pool-size constants below bound it). Capacities are means over several input realisations (twelve for the clock scan; Appendix A).

III. RESULTS

A. The product carries the memory, in quantities chemistry could use

A downstream reaction has no spectrometer; it has products. We therefore restrict the readout to observables a downstream reaction could plausibly use (whether one does is the third statement above, not modelled here), with recombination on so that the quantity read is the yield actually produced, $Y_X = k_X \int_0^T \mathrm{Tr}[P_X \rho(t)]\, dt$. Table II and Fig. 2 list all seven routes the model defines, each against its own floor.

The floors differ because of nonlinearity, not channel count: every yield is linear in the singlet fraction s of the current cycle, so a yield route of any channel count predicts s but not s^2 without memory (floor 1), whereas the product polarisations depend nonlinearly on s and give s^2 as well (floor about 2). The three-micro-environment route reads three copies of the molecule with different lifetimes in parallel. Three things follow. A single end-point

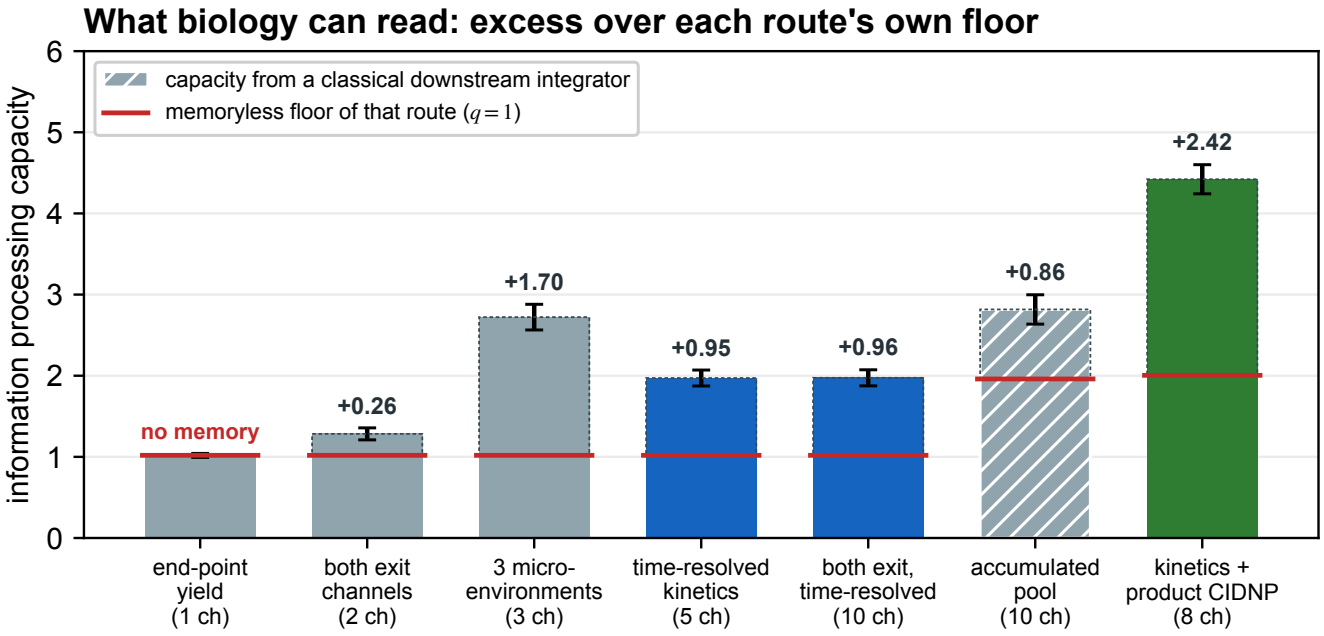


FIG. 2. **Readout is not the bottleneck.** Capacity of the chemically accessible routes at the cryptochrome point ($B = 50\ \mu$T, $J = 0$, $\tau = 1\ \mu$s). Red bars are each route's own memoryless floor; the excess printed above each bar is what measures memory. Kinetics clear their floor by 0.95, the product's nuclear polarisation (CIDNP) by 2.42, the end-point yield by nothing. The accumulated pool is hatched because a leaky integrator is itself a delay line.

TABLE II. Capacity recovered by chemically accessible observables at the cryptochrome point, each scored against its own memoryless floor (survivor register); the last column is the same route on the physically faithful product register.

readout route	ch.	IPC	floor	excess	product
end-point singlet yield	1	1.02	1.02	0.00	1.01
both exit channels, end point	2	1.28	1.02	0.26	—
three micro-environments (different τ)	3	2.72	1.02	1.70	—
time-resolved kinetics, one product pool	5	1.97	1.02	0.95	**2.77**
both exit channels, time-resolved	10	1.97	1.02	0.96	2.78
kinetics + CIDNP on the product	8	**4.42**	2.00	**2.42**	**4.96**
kinetics + accumulated downstream pool	10	2.82	1.96	0.86	—

yield remembers nothing: its capacity equals its floor. The time course of one product does, clearing its floor by 0.95, and adding the second exit channel buys only 0.003 more, so the number of product observables is not what limits the system. And the polarisation the product carries away is the best readout by a wide margin, 4.42 against a floor of 2.00. The accumulated-pool route is excluded from every claim because a leaky integrator is itself a delay line. The mechanism is spin sorting: the input lives in the two-electron correlation, and only spin-selective recombination converts it into a nuclear population difference (with recombination off the capacity is the floor, 1.02); with the reported electron–electron coupling[23] the electrons' interaction writes it into the nuclei as well (Sec. III F).

With the product register (Eq. (A4)) the same readouts are stronger: time-resolved kinetics rise from 1.97 to 2.77 and the CIDNP route from 4.42 to 4.96 (Table II, last column), so the survivor numbers reported above are the conservative ones. Every result below that *reduces* the claims has been re-run with the product register and survives (supplementary material, Sec. S4).

B. The memory is nuclear, so the clock is the turnover

This section uses the second-long nuclear T_1 measured at high field as a placeholder, to isolate the clock; Sec. III D replaces it by the geomagnetic-field value, and the numbers that decide the question are those of Table V. The horizon over which a reservoir remembers is its capacity multiplied by its cycle time. Identifying that

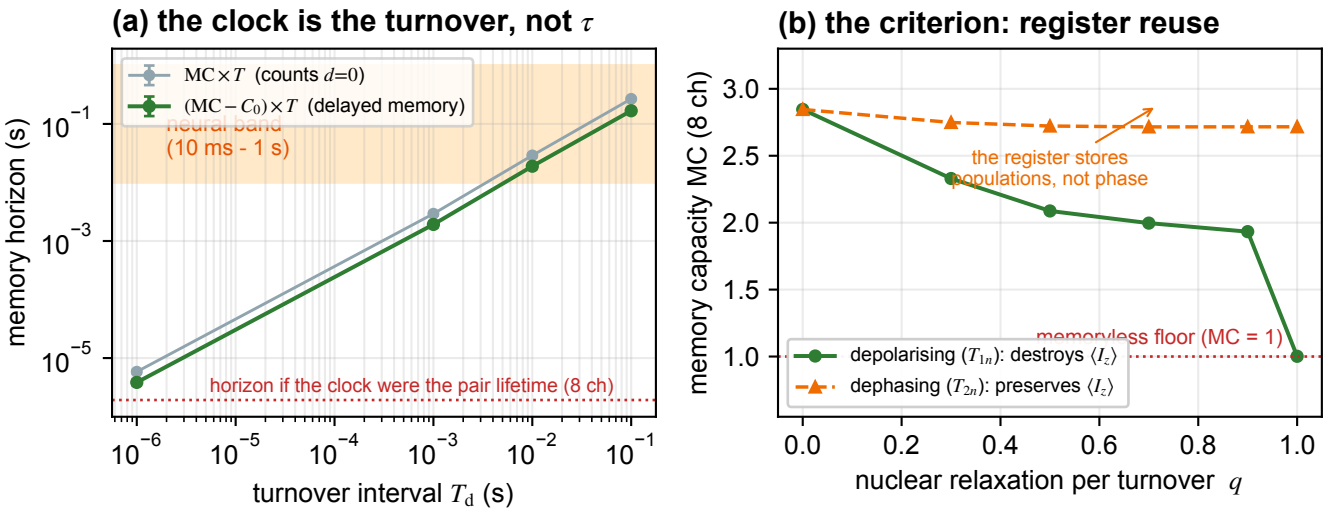


FIG. 3. **(a)** Memory horizon against turnover interval with the placeholder $T_{1n} = 1$ s (twelve realisations; error bars, between-seed s.d.); shaded, 10 ms–1 s, the range of cortical rhythms[8]; grey, raw MC × T_{turnover}; green, the delayed-memory horizon; dotted, the horizon if the clock were the pair lifetime. **(b)** Depolarising the register between turnovers with probability q (green) drives the capacity to the floor at $q = 1$; pure dephasing (orange) costs under five per cent: the register stores populations and the bounding constant is T_{1n}.

cycle time with the pair lifetime $\tau \sim 1$ μs would put the system 10^3–10^5 below neural timescales, but the memory is carried by the nuclei, which leave in the diamagnetic product and relax slowly; the cycle time is the interval between turnovers.

The horizon is quoted as $(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$, since the zero-delay term $C_0 \simeq 1$ reports the present input, and is the integrated delay capacity of Sec. II (1.82 in the six-seed run there, 1.92 in the twelve-seed clock scan of Table III); the delay profile falls below 5% of C_0 by four turnovers, so no quoted horizon hides a long tail (supplementary material, Sec. S2). Unless stated otherwise, capacities in this and the next section are linear memory capacities MC of the eight-channel CIDNP route with the survivor register, whose floor is 1.00; the IPC values of Table II include the quadratic terms and have the floor 2.00.

Inserting a pause T_{d} between turnovers and relaxing the register with probability $q = 1 - \exp(-T_{\mathrm{d}}/T_{1n})$, with the placeholder $T_{1n} = 1$ s, gives Table III and Fig. 3(a).

With the same seeds at every T_{d}, the change is assessed on paired differences: across five decades the capacity falls by 0.239 ± 0.028 (SEM, $n = 12$, $t = 8.46$), that is 8.2%, resolved well above the seed noise and the time-grid error (0.03% from $n_t = 96$ to 192) and concentrated in the last decade; a single-realisation scan cannot resolve it and is retained as supplementary material, Table S11, whose single realisation makes it a control, not a result. A 10^5-fold change in the clock moves the capacity by under a tenth: the horizon tracks the turnover interval, and a turnover of 10–100 ms places it at 19–168 ms, inside the neural band, *provided the register survives the pause*. That proviso is what the rest of the paper is about.

C. The criterion: the same register must be reused

Everything above assumes that the register written into is the *same* register on the next turnover. Depolarising the nuclei with probability q between cycles gives Fig. 3(b) (four seeds; the twelve-seed value at $q \to 0$ is 2.92): the eight-channel MC falls from 2.85 at $q = 0$ to 2.33 at $q = 0.3$, 2.09 at $q = 0.5$, 2.00 at 0.7 and 1.93 at 0.9, and at $q = 1$, a fresh molecule every cycle, to the floor MC = 1.00.

Which relaxation process enforces this can be asked directly. Replacing the depolarising channel by pure dephasing, which destroys nuclear coherence but preserves $\langle I_z \rangle$, leaves the capacity within five per cent (Table IV): the memory is carried by nuclear populations, exactly what CIDNP reports, and the relevant constant is T_{1n}. At the geomagnetic field, however, extreme narrowing ($\omega\tau_c \approx 2 \times 10^{-4}$) makes T_{1n} and T_{2n} coincide to one part in 10^7 (supplementary material, Sec. S12), so the distinction buys nothing at the operating point.

Not every nucleus can hold the register. Protons in proteins relax on the second timescale at NMR fields[34], but ^{14}N is quadrupolar (spin 1) and relaxes in $\sim$ 0.2–0.3 μs at the geomagnetic field (supplementary material, Sec. S12), so the two flavin ^{14}N are wiped every cycle. Restricting the register to the tryptophan proton leaves IPC = 2.78 against the IPC floor of 2.00, an excess of 0.78 where the full register gave 2.42, and the kinetic readout falls to its floor (supplementary material, Sec. S9): the criterion survives on a proton register, at 32% of the excess (29% with the product register) and through CIDNP alone. Scanning that register as the biology would keep it, ^{14}N wiped, product carried and the proton alone de-

TABLE III. The memory horizon against the turnover interval, twelve input realisations.

T_{d}	q	MC (8 ch)	MC $\times\, T$	(MC $- C_0$) $\times\, T$
1 μs	10^{-6}	2.919 ± 0.205	5.8 μs	3.8 μs
1 ms	10^{-3}	2.916 ± 0.204	2.9 ms	1.9 ms
10 ms	10^{-2}	2.894 ± 0.196	29 ms	19 ms
100 ms	9.5×10^{-2}	2.679 ± 0.124	268 ms	168 ms

TABLE IV. Which nuclear relaxation process enforces the criterion.

q	MC (8 ch), depolarising	MC (8 ch), dephasing
0.0	2.846	2.846
0.5	2.088	2.722
0.9	1.933	2.716
1.0	1.002	2.716

polarised with probability q_{H}, gives an MC of 1.85 at $q_{\mathrm{H}} = 0$ that holds to 1.89 at 0.5 and falls through 1.67 at 0.8 and 1.30 at 0.9 to the floor 1.00 at $q_{\mathrm{H}} = 1$. This is the curve the criterion's numbers are read from.

The criterion is then the following. *The same nuclear register must be re-used across successive turnovers, and successive turnovers must be driven by distinct input samples. The register must be carried by protons, since the flavin ^{14}N cannot hold one. Given that, the memory horizon is* (MC $- C_0$) $\times\, T_{\mathrm{turnover}}$*, bounded above by the register's own T_{1n}; without it, the system retains nothing.* It is falsifiable by chemistry: it fails if the radical pair is regenerated on a fresh molecule each turnover, or if T_{1n} in the diamagnetic product is shorter than the turnover interval.

D. The criterion's constants, predicted

Why the field matters. A nuclear spin relaxes through the fluctuating magnetic field of neighbouring nuclei, sampled at its own Larmor frequency; the fluctuation is produced by molecular tumbling with correlation time τ_c (about 15 ns for a 60 kDa protein) and by faster local motion with correlation time τ_{int}, which averages away all but a fraction S^2 of the interaction; the effective correlation time of the proton's coupling to its partner, $\tau_{\mathrm{eff}} = S^2\tau_c + (1 - S^2)\tau_{\mathrm{int}}$, is what "how fast the register reorients" means below. At 50 μT the proton Larmor frequency is 2 kHz, far below $1/\tau_c$, so the spectral density sits at its zero-frequency maximum and T_1 is as short as tumbling can make it; at 9.4 T it is sampled at 400 MHz, far above $1/\tau_c$, and T_1 is seconds. Protons of the surrounding water and protein (the intermolecular bath, calibrated at 54% of the intramolecular rate; supplementary material, Sec. S12) add to the rate.

Neither T_{turnover} nor T_{1n} has been measured for a neuronal cryptochrome, but T_{1n} can be predicted from measured constants (supplementary material, Sec. S12), and the result is adverse: the second-long T_{1n} used above is a high-field value, and a proton on a 60 kDa protein relaxes in 2.4 ms at 50 μT against 9.0 s at 9.4 T. What sets the ceiling is τ_{eff}, together with the sum of dipolar couplings to intramolecular and intermolecular partners.

Feeding $q = 1 - \exp(-T_{\mathrm{d}}/T_1)$ with the predicted T_1 into the computed MC(q) curve and maximising the horizon over T_{d} gives the ceiling on the horizon, and solving for the τ_{eff} at which the ceiling reaches 10 ms gives the boundary (Table V). Four assumptions are compared (A–D in Table V): every nucleus kept; the proton alone on the product register; the same with only the recombined third regenerated; and the same with both branches continuing. Assumption B is the main line of this paper, and every biological number in the abstract and conclusion is read from it; C and D bracket what closing the molecule budget does to it, and A is the spin-dynamical upper bound. On the register the biology can keep the cap is 2.6 ms (1.7 ms with the bath), below the 10 ms edge of the band, and the band opens only if the register reorients 3.8 (5.9) times faster than the protein; closing the budget makes it 10 (15) times. The flavin 8α-methyl protons, partly averaged by methyl rotation ($S^2 = 1/4$), relax more slowly and would clear the band on the full register but not on the proton-only one. The viable turnover intervals at a given τ_c form a window (10–14 ms at 5 ns, 5.6–790 ms at 0.1 ns, bath included), so the two conditions are not independent.

The drive cannot be light: with the measured flavin extinction coefficient a single flavin turns over once per 7.7 s in full sunlight and once per some hundreds of years inside a skull, so a photochemical cycle fails by about ten orders of magnitude and the regeneration must be catalytic. The Hamiltonian is that of the photo-generated [$\mathrm{FAD}^{\bullet-}$ $\mathrm{TrpH}^{\bullet+}$] pair, and no dark, catalytic route that regenerates that pair on a cryptochrome is established; the criterion therefore names a requirement on whatever flavoenzyme cycle is proposed to meet it, not a cycle known to exist.

E. There is no quantum advantage

A capacity is only meaningful against a classical reference read out symmetrically. The reference is a tanh echo-state network (ESN): a handful of coupled classi-

TABLE V. The criterion's numbers at the geomagnetic field: the ceiling on the memory horizon for a register that reorients with the intact protein, and the τ_{eff} below which the horizon reaches the 10 ms band edge, without and with the intermolecular bath. A assumes every nucleus is kept (the curve of Fig. 3b); B keeps the tryptophan proton on the product register; C regenerates only the third of the molecules that recombined by τ and replaces the rest with fresh ones; D lets recombined and unreacted molecules both continue, in which case their opposite polarisations largely cancel and the band is not reached.

register	ceiling (ms)		τ_{eff} (ns)	
	dry	bath	dry	bath
A: all nuclei kept, Trp Hβ	5.4	3.5	8.3	5.4
A, flavin methyl	11.2	—	17.1	9.6
B: proton (product), Trp Hβ	2.6	1.7	4.0	2.6
B, flavin methyl	5.4	3.0	8.2	4.6
C: as B, recombined third only	1.0	0.7	1.6	1.0
D: as B, both branches	none	none	none	none

cal variables, each updated from its neighbours and the current input, read with the same linear fit. At the engineered point, with a twelve-observable spin readout (IPC $= 5.6 \pm 0.2$), the reservoir is modestly ahead of a network matched by element count and well behind one matched by readout count (supplementary material, Sec. S6); which match is fair depends on a cost model we do not have.

At the cryptochrome point the comparison is channel-for-channel against the chemical readouts (Table VI). The fit penalises large weights by a ridge term λ. With $\lambda = 10^{-6}$ fixed on the raw readout values the network wins by 57% at eight channels and 127% at five, but that margin belongs to the regulariser: the chemical channels have amplitudes of order 10^{-2} and the network states of order one, so one absolute ridge penalises the quantum readout far more. Rescaling each readout value to unit variance and choosing λ for each system by its own out-of-sample error (cross-validation) removes it: the paired difference is $+0.06 \pm 0.16$ (SEM, $n = 12$) at eight channels and -0.10 ± 0.10 at five, that is $+1\%$ and -2%, neither resolved from zero. On no accounting does the radical pair compute better than a small classical network, and on no properly regularised accounting does it compute worse. The statement is bounded by its task and comparison: reconstruction of past inputs and their quadratic functions, against a network given the same number of readout values; it does not show that quantum processes play no role in the capacity. A correlated-spin reservoir with full quantum readout has been reported to outperform classical reservoirs[46]; there is no tension, because what is read out differs. The advantage, where it exists, lives in the readout access, which is exactly what a chemical cycle does not grant.

How much of the capacity is quantum at all is addressed in the supplementary material (Secs. S5 and S8)

TABLE VI. The same comparison at the cryptochrome point, channel-for-channel. Left: ridge fixed at $\lambda = 10^{-6}$ on raw features (six seeds). Right: features standardised and λ chosen by five-fold cross-validation on the training half, for both systems (twelve paired seeds).

system (readout features)	λ fixed	λ by CV
quantum, kinetics (5 ch)	1.97 ± 0.10	4.64 ± 0.29
ESN, 5 nodes (5 features)	4.47 ± 0.33	4.55 ± 0.30
quantum, +CIDNP (8 ch)	4.42 ± 0.18	6.90 ± 0.47
ESN, 8 nodes (8 features)	6.94 ± 0.56	6.95 ± 0.61
ESN, 12 nodes (12 features)	9.62 ± 0.93	—

and is not needed for the criterion.

F. What the numbers do and do not depend on

Five sensitivities bear on the criterion; each is computed in full in the supplementary material.

The coupling premise. The cryptochrome point idealises "separated" as no electron–electron coupling, whereas the source reports $D = -11.4$ MHz and an exchange gap of 1.3–126 MHz of undetermined sign[23]. With those values (supplementary material, Sec. S14) the no-recombination capacity rises from the floor to 5.1–5.9 and the chemically accessible routes move as Table S31 records, with every floor unchanged; the ceiling rises from 3.4 to 5.7 ms with the bath, still below the band. The criterion survives the premise; its margin is what the premise sets.

The hyperfine tensors. With the published tensor set of the pair[12,21], orientation-averaged and with a fourth nucleus added (supplementary material, Sec. S11), the full register's CIDNP excess (product register, IPC) falls from 2.96 to 0.93, while the proton-only register the criterion rests on retains 0.60 against 0.86 (0.98–1.47 with a second proton); the headline capacities overstate the full register, the criterion's numbers are not loosened.

Regularisation and counting noise. The fixed ridge is load-bearing for the kinetic route (capacity 4.50 to 1.00 as λ runs from 10^{-12} to 10^{-1}), so the reported values are conservative with respect to it but not independent of it; the criterion's curve $\mathrm{MC}(q)$ is computed at the fixed ridge, and since the cross-validated readout raises the quantum excess by at most a factor of 1.6 (Table VI), it could loosen the boundary by at most that factor, against the 4–15 the register must gain. The physically meaningful noise is that of counting a pool of N molecules: a polarisation of order 10^{-2} is the mean of Ny_S values of ± 1, with variance about $1/Ny_S$. Counted that way (supplementary material, Sec. S7), the CIDNP excess is 2.41 of its noiseless 2.42 at $N = 10^{10}$, 1.93 at 10^8 and 0.49 at 10^6: we call the readout counting-noise limited where the excess has fallen more than 3% below its noiseless value, which is below about 10^9 molecules per pool, a pool being

the molecules that share one input and whose yields and polarisations are summed into one readout; the criterion carries that number as a third constant.

What carries the input. No chemistry is named that modulates s. Driving the reservoir through the field magnitude over 25–75 μT gives a CIDNP-route excess (IPC, product register) of 3.14, against 2.96 with the singlet fraction as input, and through the recombination rate 2.42 (supplementary material, Sec. S3), so the criterion does not rest on a chemically unmotivated input. Pooling $\sim 10^{10}$ copies is safe against heterogeneity but not against desynchronisation, the copies' cycles falling out of step, which CIDNP alone survives (Sec. S10).

IV. DISCUSSION

What the calculation shows. A radical pair regenerated on the same molecule has a state variable that outlives each cycle: the polarisation of the nuclei that leave in its product. That variable is written by spin-selective recombination, read by CIDNP and erased by nuclear spin–lattice relaxation, so the memory of a cycled radical-pair reaction is set by three constants that belong to the chemistry and not to the pair's coherence: the turnover interval, the product's proton T_1 at the working field, and the number of molecules pooled into one readout. The pair's microsecond lifetime, usually taken as the timescale of radical-pair effects, does not enter. At the geomagnetic field the second constant is milliseconds, and that is what limits the memory.

What would settle it. The criterion is addressable in vitro before it is addressable in a neuron. Covalently linked donor–bridge–acceptor radical pairs have tunable exchange coupling and lifetime and are read by transient EPR and optical or CIDNP spectroscopy[2,14,41]; cycling such a system repeatedly and asking whether the product yield of turnover k depends on the input at turnover $k-1$, $k-2$, … measures the memory horizon directly, and varying the interval tests the criterion where it is sharpest. The product's proton T_1 at 50 μT is measurable by relaxation dispersion, and the turnover interval of a candidate flavoenzyme cycle by steady-state kinetics.

What it implies for radical-pair computation in neurons. Two objections commonly raised against the conjecture fail, though not as its proponents would wish: a chemical readout is not too impoverished, because it need not be scalar, and the microsecond lifetime does not put neural timescales out of reach, because it is the wrong clock. The limit is the product's nuclear relaxation, and it is severe. Reaching the 10 ms edge of the band of neural rhythms requires a proton register that reorients four to fifteen times faster than the protein tumbles, regeneration of the pair on the same molecule by a catalytic rather than photochemical cycle, a pool of about 10^9 molecules sharing one input, and a downstream reaction that reads nuclear polarisation. None of these is established for a neuronal protein, and the few-spin system computes no better than a classical network of the same size. Whether a neuronal cryptochrome forms a separated radical pair at all is unestablished, and a behavioural test of magnetic-field effects in *Drosophila* found none[55]. The criterion therefore does not support the conjecture; it states what a proposal would have to demonstrate, in constants that can be measured before any neuron is involved.

Relation to prior work. Quantum reservoir computing in spin ensembles is established[13,26–29], with dissipation as a resource[42], memory kept outside the coherent state[43,44], input-rate-dependent value of entanglement[45] and, with full quantum readout, an advantage over large classical reservoirs[46]; here the readout is what chemistry can carry away and the memory sits in a nuclear register. Wong *et al.*[35] asked whether nuclear polarisation accumulates across cryptochrome photocycles and found that an idealised gain does not survive conditions approaching those in vivo; our criterion states what would have to be true for it to survive. Photo-CIDNP has been observed in flavoproteins[40,53,54] though not in cryptochrome. The decoherence objection to quantum-brain proposals[9] concerns ion positions, not nuclear spins; the objection that applies here is nuclear relaxation in the product, whose record is discouraging[6,33,36]. Thermal modulation of the hyperfine tensors[47,48], decoherence-enhanced sensing[49], Zeno-protected tightly coupled pairs[50], the estimated FAD–TrpH couplings[51] and sixty-spin tensor-network propagation[52] each bound what a fixed-tensor, five-spin model can say, and are the reasons the premise and the tensors are computed in Sec. III F rather than assumed.

Limitations. The model is a five-spin reduction with spin-1/2 nitrogen; the tryptophan nuclei beyond Hβ1 and H1 are not treated, only degree-2 nonlinear capacity is counted, the headline capacities overstate the full register under the published tensors, and the closed cycle assumes that molecules unreacted at τ hand on the nuclear state they then have. No chemistry that modulates the input is identified and no dark, catalytic cycle regenerating this pair is known; the criterion states requirements, not a mechanism.

V. CONCLUSION

A radical pair regenerated on the same molecule remembers its earlier turnovers in the nuclear polarisation of its product. The memory survives recombination, so its clock is the turnover interval, and the product's polarisation reads it out at a capacity of 4.4 against a floor of 2.0 with no spectroscopy; an end-point yield reads none of it. It is held by protons, whose relaxation at the geomagnetic field, predicted rather than assumed, caps the memory horizon at 1.7–2.6 ms and at about 1 ms once the cycle's molecule budget is closed. The readout is counting-noise limited below about 10^9 molecules per pool, and a matched classical network does as well un-

der a fair regularisation. For the conjecture that such chemistry computes in neurons this is a bound: the neural band is reached only if the register reorients four to fifteen times faster than the protein and the drive is catalytic. All three constants of the criterion are measurable in vitro.

SUPPLEMENTARY MATERIAL

See the supplementary material for the Liouville-space solver (S1), the capacity estimator and its controls (S2), the injection map (S3), the readout routes (S4), the coherence fraction (S5), the classical baselines (S6), the turnover clock and criterion (S7), the semiclassical reference (S8), which nuclei hold the register (S9), ensemble averaging (S10), anisotropic hyperfine coupling (S11), the predicted constants (S12), spin parameters (S13), the electron–electron coupling (S14), the reservoir protocol (S15) and generative AI use (S16).

DATA ACCESSIBILITY

All analyses are in the open `qbscreen` package, `https://github.com/deeptell-inc/new_Q_brain`, with outputs in `simulation_results/`. The capacity bound and the solver's agreement with the closed-form coherent limit are checked by the suite in `qbscreen/tests/` (315 tests, all passing). Thirty-seven tests in `test_panel_claims.py` bind the headline quantities and load-bearing claims, and a further one hundred and forty-four in `test_table_rows.py` bind the capacity and standard-deviation cells of every table in the main text and the supplementary material, in both directions, so the suite fails if either side moves. The LaTeX sources those tests read accompany the submission and are not in the public repository, where they skip.

AUTHORS' CONTRIBUTIONS

H.W. conceived the study, developed the master-equation and reservoir code, ran the simulations, and wrote the manuscript. T.T. contributed to the spin-dynamics formulation and the analysis of the nuclear-register criterion. Both authors reviewed the results and approved the manuscript.

COMPETING INTERESTS

We declare we have no competing interests.

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USE OF GENERATIVE AI TOOLS

Claude Opus 5, Opus 5.2 and Fable 5.1 (Anthropic) and GPT-5.6 (OpenAI) were used, under the authors' direction and review, to write the `qbscreen` package, to draft the text from the authors' outlines and results, and to review the analysis adversarially (supplementary material, Sec. S16). No AI tool is an author; the authors take full responsibility for the content.

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All numerical results are reproducible from the open-source `qbscreen` package.

Appendix A: Model and methods

1. Reservoir Hamiltonian

The reservoir is $N = 5$ spins, two radical-pair electrons (e_1, e_2) and three spin-1/2 nuclei, with Hamiltonian (in MHz)

$$H = \omega_e \left(S_z^{e_1} + S_z^{e_2}\right) + \sum_{\mu=x,y,z} \Big[A_P \, S_\mu^{e_1} I_\mu^{P} + A_{H_1} S_\mu^{e_1} I_\mu^{H_1} + A_{e_2} S_\mu^{e_2} I_\mu^{H_2} + J \, S_\mu^{e_1} S_\mu^{e_2} \Big], \tag{A1}$$

$\omega_e = g_e \mu_B B/h$. At the cryptochrome point the nuclei carry the couplings of ^{14}N5 and ^{14}N10 on FAD$^{\bullet-}$ and Hβ1 on TrpH$^{\bullet+}$ (supplementary material, Table S29), $B = 50$ μT, $\tau = T_2^e = 1$ μs and $J = 0$ as the reference premise; Sec. III F reruns every quantity with the reported exchange and dipolar couplings[23] (supplementary material, Sec. S14). At the engineered point we use $A_{e_1 a} = 80$, $A_{e_1 b} = 40$, $A_{e_2 a} = 15$ MHz, $J = 2$ MHz and $B = 1$ mT (supplementary material, Sec. S15).

2. Master equation and propagator

Each reservoir update is the completely positive map $\rho \mapsto e^{\mathcal{L}\tau}\rho$ on the unreacted pair, trace-preserving when recombination is off and trace-decreasing when it is on (the lost trace being the product, accounted for by Eq. (A4)), generated by the Haberkorn master equation[17,30] with electronic pure dephasing,

$$\frac{d\rho}{dt} = -i[H,\rho] - \tfrac{1}{2}\{k_S P_S + k_T P_T, \rho\} + \sum_i \frac{2}{T_2^e}\left(S_z^i \rho S_z^i - \tfrac{1}{4}\rho\right), \tag{A2}$$

solved exactly in Liouville space; for $1/T_2^e \to 0$, $k_S = k_T$ it reproduces the closed-form coherent singlet yield to better than 10^{-9} (supplementary material, Sec. S1).

Every chemical-readout, clock and criterion result uses $k_S = 1$, $k_T = 0.2\ \mu\mathrm{s}^{-1}$.

3. Reservoir protocol

At step k the nuclei are retained and the electron pair is recreated as a spin-correlated pair whose singlet character encodes the input,

$$\rho_e(s_k) = s_k\,|S\rangle\langle S| + (1 - s_k)\,P_T/3, \qquad \text{(A3)}$$

so that the state entering cycle k is $\rho_e(s_k) \otimes \sigma_{\mathrm{nuc}}^{(k)}$; it evolves for τ under $e^{\mathcal{L}\tau}$, the observables are read out, and a linear ridge readout ($\lambda = 10^{-6}$ unless stated) is trained on them. Inputs s_k are i.i.d. uniform on $[0, 1]$; the first 80–100 steps are discarded as washout, so the capacities are independent of the initial spin state. The survivor register hands on the renormalised nuclear state of the pairs that had not recombined by τ; the product register hands on the nuclear state carried away by the recombined molecules,

$$\sigma_{\mathrm{nuc}} \propto \int_0^\tau \left[k_S\,\mathrm{Tr}_e(P_S\rho P_S) + k_T\,\mathrm{Tr}_e(P_T\rho P_T)\right]\mathrm{d}t, \quad \text{(A4)}$$

with the electron pair reborn unpolarised; the closed carrier sums both branches with their molecule-number weights. The number of input realisations is not uniform and is stated with each result: 12 for the turnover-clock scan and its paired test, which carry the central claim; 8 for the classical baselines and the T_2^e scan at the engineered point; 6 for the headline capacity, the chemical-readout survey, the route floors, the delay kernel, the grid scans, the nuclide scan and the cryptochrome-point classical baselines; 4 for the register-reuse and nuclear-channel scans and the semiclassical reference; and 3 for the ensemble and anisotropy scans. Four quantities are single-realisation: the ridge and shot-noise sensitivity scans and the semiclassical integration-step check, which compare variants on one fixed input by design, and a single-realisation clock scan retained as a control. Operating points and convergence checks are in supplementary material Sec. S15.

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